

Slide 1: Title

AI-Powered Candidate Discovery & Ranking System

Data & AI Challenge 2026

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Technologies: Python, Sentence Transformers, Scikit-Learn, Pandas

Slide 2: Problem Statement

Traditional hiring systems rely heavily on keyword matching.

Challenges:

- Good candidates are often missed.
- Keyword filters do not understand context.
- Recruiters spend significant time manually screening profiles.

Objective:

Build an AI-based ranking system that understands job requirements and recommends the most relevant candidates.

Slide 3: Dataset Overview

Dataset Components:

- Candidate Profiles
- Career History
- Skills Information
- Education Details
- Certifications
- Redrob Behavioral Signals

Total Candidate Records: ~100,000+

Output:

Rank candidates according to their suitability for a given job description.

Slide 4: Candidate Schema

Each candidate contains:

Profile: • Headline • Summary • Current Title • Experience

Career History: • Previous Roles • Responsibilities • Achievements

Skills: • Skill Name • Proficiency • Endorsements

Behavioral Signals: • Open To Work • Recruiter Response Rate • GitHub Activity • Notice Period • Relocation Preference

Slide 5: Semantic Search Architecture

Step 1: Convert Job Description into vector embedding.

Step 2: Convert Candidate Profile + Career History into vector embeddings.

Step 3: Calculate semantic similarity using cosine similarity.

Model Used:

all-MiniLM-L6-v2

Benefit:

Understands meaning rather than exact keyword matching.

Slide 6: Skills Matching Logic

Important AI Skills Considered:

• Python • NLP • LLM • LoRA • Retrieval Systems • Ranking Systems • Milvus • Pinecone • FAISS • Embeddings

Skills Score:

Matched Skills / Total Required Skills

This rewards candidates possessing relevant AI expertise.

Slide 7: Behavioral Signal Scoring

Behavioral indicators improve ranking quality.

Signals Used:

• Open To Work • Recruiter Response Rate • GitHub Activity Score • Notice Period • Willingness To Relocate

Why?

These signals indicate candidate availability, engagement, and hiring readiness.

Slide 8: Final Ranking Formula

Final Score =

50% Semantic Match + 20% Skills Match + 15% Experience Match + 15% Behavioral Score

Benefits:

• Context-aware ranking • Skills-based evaluation • Recruiter-friendly recommendations

Slide 9: Results & Top Candidates

Example Top Ranked Candidates

1. CAND_0011162 – Score: 0.6727
2. CAND_0000031 – Score: 0.6513
3. CAND_0018499 – Score: 0.6472
4. CAND_0002025 – Score: 0.6445
5. CAND_0018549 – Score: 0.6225

Generated Outputs:

• ranked_candidates.csv • sample_submission.csv

Slide 10: Future Improvements

Future Enhancements:

• Cross-Encoder Re-ranking • Large Language Model Evaluation • Dynamic Skill Extraction • Vector Database Integration • Real-Time Recruiter Feedback Loop • Personalized Candidate Recommendations

Impact:

More accurate, explainable, and scalable hiring decisions.

Thank You